

Number System

1. Complete the table with equivalents Binary Octal & hexa for: 143.75 (decimal), 11011.101 (binary), 453 (octal), $1F.C$ (hex)

Decimal	Binary	Octal	Hexa
143.75	10001111.11	217.6	$8F.C$
27.625	11011.101	33.5	$1B.A$
299	100101011	453	$12B$
31.75	00011111.1100	37.60	$1F.C$

2. Find Equivalents.

a) $324_5 = ?_{10}$
 $3 \times 5^2 + 2 \times 5 + 4 = (89)_{10}$

b) $32.23_4 = ?_{10}$
 $(3 \times 4) + 2 + 2/4 + 3/16$
 $= 14.6875_{10}$

c) $231_{10} = ?_7$

$$\begin{array}{r|l} 7 & 231 - 0 \\ \hline 7 & 33 - 5 \\ \hline & 4 \end{array} \quad 450_7$$

d) $(189)_{10} = ?_6$

$$\begin{array}{r|l} 6 & 189 - 3 \\ \hline 6 & 31 - 1 \\ \hline & 5 \end{array} = (513)_6$$

3. Find the radix

(a) $\left(\frac{312}{20}\right)_r = (13.1)_r$

$$= \frac{(312)_r}{(20)_r} = 13.1 = \frac{3r^2 + r + 2}{2r + 0} = r + 3 + r^{-1}$$

$$= 3r^2 + r + 2 = 2r^2 + 6r + 2.$$

$$r^2 - 5r = 0$$

$$r(r-5) = 0 \quad \underline{r=5}$$

$$(b) \quad 26r + 2136r = 433r.$$

$$2r^2 + 6r + 4 + r^2 + 3r + 6 = 4r^2 + 3r + 3$$

$$3r^2 + 9r + 10 - 4r^2 - 3r - 3 = 0$$

$$-r^2 + 6r + 7 = 0$$

$$r^2 - 6r - 7 = 0$$

$$(r-7)(r+1) = 0$$

$$r = 7.$$

$$(c) \quad 221505_8 = 12345r.$$

$$(2 \times 8^5) + (2 \times 8^4) + (1 \times 8^3) + (5 \times 8^2)$$

$$= r^4 + (2 \times r^3) + (3 \times r^2) + (4 \times r) + 5$$

$$= 74564 - r^4 + 2r^3 + 3r^2 + 4r + 5$$

$$\text{By solving } \underline{\underline{r=16}}$$

4. Addition.

$$(a) \quad 1101_2 + 11011111_2$$

$$\begin{array}{r} 11011111 \\ 00001101 \\ \hline 11101100_2 \end{array}$$

$$(b) \quad 77652_8 + 76543_8$$

$$\begin{array}{r} 77652 \\ 76543 \\ \hline 176415_{(8)} \end{array}$$

$$10) 578A_{16} + ABDE_{16}$$

$$578A$$

$$ABDE$$

$$\hline 10368$$

$$(a) 234_6 + 315_6$$

$$234$$

$$+ 315$$

$$\hline 353$$

5) Binary to gray code

$$(a) 10101_2$$

$$\text{Gray} = 11111$$

$$1 \oplus 0 = 1$$

$$0 \oplus 1 = 1$$

$$1 \oplus 0 = 1$$

$$0 \oplus 1 = 1$$

$$(b) 01110_2$$

$$\text{Gray} = 01001$$

6) Gray code to binary

$$a) 01101$$

$$\text{Binary } (01001)_2$$

$$0 \oplus 1 = 1$$

$$1 \oplus 1 = 0$$

$$0 \oplus 0 = 0$$

$$0 \oplus 1 = 1$$

$$b) 10101$$

$$\text{Binary} = 11001$$